You Asked

1. **Compulsory question**

(a) State True or False: (1×5=5)

(i) The concept of Biodiversity Hotspots was introduced by Norman Myers.

(ii) Montreal Protocol is an international agreement to phase out the production and consumption of Ozone Depleting Substances.

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(iii) Delhi lies in Trans-Himalayan biogeographic zone of India.

(iv) Royal Bengal Tiger is a flagship species of Indian wildlife.

(v) Use of fossil fuels helps in lowering carbon footprint.

Kimi

Here are the answers to the True/False questions:

**(i) The concept of Biodiversity Hotspots was introduced by Norman Myers. True** — Norman Myers introduced the concept of biodiversity hotspots in 1988.

**(ii) Montreal Protocol is an international agreement to phase out the production and consumption of Ozone Depleting Substances. True** — The Montreal Protocol (1987) is indeed an international treaty designed to protect the ozone layer by phasing out the production and consumption of ozone-depleting substances (ODS).

**(iii) Delhi lies in Trans-Himalayan biogeographic zone of India. False** — Delhi does not lie in the Trans-Himalayan biogeographic zone. The Trans-Himalayan zone includes areas like Ladakh and parts of Jammu & Kashmir. Delhi falls in the **Gangetic Plain** biogeographic zone.

**(iv) Royal Bengal Tiger is a flagship species of Indian wildlife. True** — The Royal Bengal Tiger (*Panthera tigris tigris*) is indeed a flagship species of Indian wildlife and is the national animal of India.

**(v) Use of fossil fuels helps in lowering carbon footprint. False** — The use of fossil fuels (coal, oil, natural gas) **increases** carbon footprint because they release carbon dioxide ($CO_2$) and other greenhouse gases when burned, contributing to climate change. Lowering carbon footprint requires reducing fossil fuel use and switching to renewable energy sources.

You Asked

(b) Fill in the blanks : (1×5=5)

(i) ______ is an example of invasive plant species in India.

(ii) One of the foremost reasons for biodiversity loss across the world is ______ .

(iii) ______ is one of the prominent people associated with Chipko movement.

(iv) IUCN has categorized various species into ______ (number) conservation status categories

P.T.O.

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(v) The National Action Plan for Climate Change (NAPCC) was launched in the year ______

Kimi

Here are the answers to the fill-in-the-blanks:

**(i) Lantana camara** (or *Parthenium hysterophorus* / Water hyacinth / *Prosopis juliflora*) is an example of invasive plant species in India.

**(ii)** One of the foremost reasons for biodiversity loss across the world is **habitat destruction** (or habitat loss / deforestation).

**(iii) Sunderlal Bahuguna** (or Chandi Prasad Bhatt / Gaura Devi) is one of the prominent people associated with the Chipko movement.

**(iv)** IUCN has categorized various species into **9** conservation status categories.

**(v)** The National Action Plan for Climate Change (NAPCC) was launched in the year **2008**.

You Asked

2. Differentiate between **any two** of the following : (2×5=10)

(a) Climate Change vs Ozone layer depletion

P.T.O.

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(b) Umbrella species vs Indicator species

(c) Silent valley movement vs Naramada Bachao Andolan

Kimi

Here are the differentiations for all three pairs — you may choose any two for your answer:

---

## (a) Climate Change vs Ozone Layer Depletion

| Basis | Climate Change | Ozone Layer Depletion |
|---|---|---|
| **Definition** | Long-term shift in global/regional climate patterns, especially a rise in average temperatures. | Thinning of the ozone layer ($O_3$) in the stratosphere, allowing more UV radiation to reach Earth. |
| **Primary Cause** | Greenhouse gases ($CO_2$, $CH_4$, $N_2O$) from burning fossil fuels, deforestation, agriculture. | Chlorofluorocarbons (CFCs), halons, and other ozone-depleting substances (ODS). |
| **Affected Region** | Global — affects the entire planet's temperature and weather systems. | Most severe over polar regions (Antarctic "ozone hole"). |
| **Key Impact** | Global warming, sea-level rise, extreme weather, melting glaciers. | Increased UV-B radiation causing skin cancer, cataracts, and harm to marine ecosystems. |
| **International Agreement** | Paris Agreement (2015), Kyoto Protocol (1997). | Montreal Protocol (1987). |

---

## (b) Umbrella Species vs Indicator Species

| Basis | Umbrella Species | Indicator Species |
|---|---|---|
| **Definition** | A species whose conservation indirectly protects many other species in the same habitat. | A species whose presence, absence, or abundance indicates the health/quality of an ecosystem. |
| **Main Purpose** | Habitat conservation — protecting their large range benefits entire ecosystems. | Environmental monitoring — they signal changes in environmental conditions. |

| Basis | Umbrella Species | Indicator Species |
|---|---|---|
| **Example** | Tiger, Elephant (protect large forest areas). | Lichens (air quality), Frogs (water quality), Mayflies (river health). |
| **Conservation Focus** | Used to design protected areas and reserve networks. | Used to assess pollution, habitat degradation, or ecosystem stress. |
| **Size/Range** | Usually large-ranging, charismatic megafauna. | Can be any organism sensitive to environmental changes. |

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### (c) Silent Valley Movement vs Narmada Bachao Andolan

| Basis | Silent Valley Movement | Narmada Bachao Andolan (NBA) |
|---|---|---|
| **Location** | Silent Valley, Kerala (Western Ghats). | Narmada River Valley (Gujarat, Madhya Pradesh, Maharashtra). |
| **Period** | 1970s–1983. | 1985–present (peak in late 1980s–1990s). |
| **Main Issue** | Proposed hydroelectric dam (Silent Valley Hydroelectric Project) threatening pristine tropical rainforest. | Construction of large dams (Sardar Sarovar Dam) causing displacement of tribals and environmental damage. |
| **Key Leader** | Sugathakumari, K. M. Chinnappa, and environmental groups. | Medha Patkar, Baba Amte, Arundhati Roy. |
| **Outcome** | Project was abandoned in 1983; Silent Valley declared a National Park in 1984. | Ongoing struggle; led to World Bank withdrawal (1993), but dam was eventually completed. |
| **Focus** | Primarily ecological/biodiversity conservation (protection of endemic species like the Lion-tailed Macaque). | Primarily human rights and rehabilitation of displaced people (tribals, farmers) alongside environmental concerns. |

You Asked

3. Briefly explain **any four** of the following : (4×2.5=10)

(a) Biosphere Reserves

(b) National Solar Mission

(c) Carbon footprint

(d) In-situ conservation strategies

(e) Role of Eco-clubs in Environmental Awareness and Conservation

Kimi

Here are brief explanations for all five topics — you may choose any four:

---

### (a) Biosphere Reserves

Biosphere Reserves are protected areas designated by UNESCO under the Man and the Biosphere (MAB) Programme. They aim to conserve biodiversity while promoting sustainable development. A biosphere reserve is divided into three zones:

- **Core zone** — strictly protected, no human activity.
- **Buffer zone** — limited human activity like research and education.
- **Transition zone** — sustainable human settlements and economic activities allowed.

India has 18 biosphere reserves (e.g., Nilgiri, Sundarbans, Nanda Devi, Pachmarhi). They serve as sites for conservation, research, and learning to reconcile biodiversity protection with human welfare.

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### (b) National Solar Mission

The **Jawaharlal Nehru National Solar Mission (JNNSM)** was launched in 2010 as one of the eight missions under India's National Action Plan on Climate Change (NAPCC). Its goal is to promote ecologically sustainable growth while addressing India's energy security challenges.

**Key objectives:**

- To establish India as a global leader in solar energy.
- To create favorable policy conditions for solar technology diffusion across the country.
- Initial target: 20 GW of grid-connected solar power by 2022 (later revised to 100 GW).

It focuses on both **solar thermal** and **solar photovoltaic** technologies, encouraging domestic manufacturing and reducing dependence on fossil fuels.

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### (c) Carbon Footprint

Carbon footprint refers to the **total amount of greenhouse gases (GHGs)** — primarily carbon dioxide ($CO_2$) — emitted directly or indirectly by an individual, organization, event, or product throughout its lifecycle. It is usually measured in **tonnes of $CO_2$ equivalent ($tCO_2e$)**.

**Sources include:**

- Burning fossil fuels for energy and transport.
- Industrial processes and agriculture.
- Deforestation and land-use changes.

Reducing carbon footprint involves using renewable energy, improving energy efficiency, afforestation, sustainable transportation, and adopting low-carbon lifestyles. It is a key metric in climate change mitigation efforts.

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### (d) In-situ Conservation Strategies

In-situ conservation means the **conservation of species in their natural habitats** without removing them from the wild. It helps maintain natural ecosystems and evolutionary processes.

**Key strategies include:**

- **Protected Areas:** National Parks, Wildlife Sanctuaries, Biosphere Reserves.
- **Sacred Groves:** Community-protected forest patches with religious significance.
- **Biodiversity Hotspots:** Regions with high species richness and endemism prioritized for conservation.
- **Community-based conservation:** Involving local communities in habitat protection.

In-situ conservation is preferred over ex-situ (off-site) methods because it preserves entire ecosystems, genetic diversity, and species interactions.

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### (e) Role of Eco-clubs in Environmental Awareness and Conservation

Eco-clubs are school- or college-based groups formed under programmes like the **National Green Corps (NGC)** by the Ministry of Environment, Forest and Climate Change (MoEFCC).

**Their roles include:**

- **Awareness:** Educating students and communities about environmental issues like pollution, climate change, and biodiversity loss.
- **Action:** Organizing tree plantations, cleanliness drives, waste management, and water conservation activities.
- **Leadership:** Developing young environmental leaders who influence family and community behavior.
- **Monitoring:** Participating in local environmental monitoring (e.g., water quality, bird counts).

Eco-clubs bridge the gap between classroom learning and real-world environmental action, fostering a culture of sustainability from an early age.

You Asked

4. Human-wildlife conflict has emerged as a major threat to biodiversity in India. Identify the possible reasons for the rise in such conflicts and suggest tangible solutions to mitigate the same? (10)

Kimi

Here is a comprehensive answer to the question on **Human-Wildlife Conflict in India**:

---

## Human-Wildlife Conflict in India: Causes and Solutions

### Introduction

Human-wildlife conflict (HWC) refers to situations where wildlife requirements encroach on those of human populations, with substantial costs to both. In India, with its high human population density and rich biodiversity, HWC has become a major threat to both biodiversity conservation and human livelihoods.

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## Reasons for the Rise in Human-Wildlife Conflict

### 1. Habitat Loss and Fragmentation

- Rapid deforestation, urbanization, and agricultural expansion have reduced and fragmented wildlife habitats.
- Linear infrastructure (roads, railways, canals) cuts through corridors, forcing animals into human settlements.
- Example: Elephant corridors in Karnataka and Kerala are increasingly blocked by plantations and settlements.

### 2. Human Encroachment into Forest Areas

- Growing population pressure has led to settlements and farming near forest boundaries.
- Tribal and rural communities depend on forest resources (fuelwood, fodder, non-timber forest products), increasing encounters with wildlife.

### 3. Decline of Natural Prey Base

- Overgrazing, poaching, and habitat degradation reduce prey availability for carnivores.
- This forces predators like tigers, leopards, and wolves to hunt livestock and occasionally enter villages.

### 4. Crop-Raiding by Herbivores

- Elephants, wild boars, nilgai, and monkeys frequently raid agricultural fields.
- Sugarcane, paddy, and maize crops are particularly attractive to wildlife.
- This causes significant economic losses to farmers and triggers retaliatory killings.

### 5. Livestock Predation

- Leopards, wolves, snow leopards, and tigers prey on cattle, goats, and sheep.
- Poor livestock management (grazing inside forests, open sheds) increases vulnerability.

### 6. Changing Land Use Patterns

- Conversion of multi-use landscapes to monoculture plantations (tea, coffee, rubber) reduces habitat quality.
- Wetland drainage and river damming alter ecosystems, displacing species.

### 7. Climate Change

- Erratic rainfall, droughts, and changing vegetation patterns push wildlife to new areas in search of food and water.
- Human-wildlife interface expands as both compete for scarce resources.

### 8. Lack of Effective Waste Management

- Improper disposal of food waste near forest edges attracts wildlife (especially leopards, bears, and monkeys) to human settlements.

### 9. Inadequate Mitigation Measures

- Poorly designed mitigation structures (fences, trenches) are easily breached by elephants.
- Lack of early warning systems and rapid response teams.

---

## Tangible Solutions to Mitigate Human-Wildlife Conflict

### 1. Habitat Restoration and Corridor Protection

- Restore degraded habitats and protect existing wildlife corridors.
- Create buffer zones around protected areas with regulated land use.
- Example: The **Elephant Corridors of India** project by WTI and the government.

### 2. Early Warning Systems

- Deploy **SMS/APP-based alert systems** (e.g., *Haathi Mere Saathi* in Assam, *WildSeve* in Karnataka).
- Use camera traps, drones, and AI-based monitoring to track animal movement and alert communities.

### 3. Physical Barriers

- Construct **solar-powered electric fences**, **elephant-proof trenches**, and **railway underpasses/overpasses**.
- Ensure community participation in construction and maintenance.

### 4. Alternative Livelihoods and Compensation

- Provide **ex-gratia compensation** for crop damage and livestock loss promptly and transparently.
- Promote **eco-tourism**, **beekeeping**, and **handicrafts** as alternative income sources near forests.

### 5. Community-Based Conservation

- Form **Van Suraksha Samitis** (Forest Protection Committees) and involve locals in conflict management.
- Train **village-level response teams** to handle wildlife encounters safely.

### 6. Crop and Livestock Management

- Promote **crop diversification** with less palatable crops (e.g., chili, ginger, citrus) near forest edges.
- Encourage **predator-proof livestock enclosures** (bomas) and night penning.

### 7. Waste Management

- Enforce strict waste disposal protocols in villages near forests.
- Use **bear-proof bins** and avoid dumping organic waste near forest boundaries.

### 8. Translocation and Rescue Operations

- Relocate problem animals (as a last resort) using trained wildlife rescue teams.
- Establish **rescue and rehabilitation centers** near conflict hotspots.

### 9. Insurance Schemes

- Introduce **livestock and crop insurance schemes** specifically covering wildlife damage.
- Example: The **Gudia Baxi** scheme in Odisha for crop loss due to elephants.

### 10. Education and Awareness

- Conduct community awareness programmes on co-existence.
- Use school eco-clubs, local media, and folk performances to spread awareness.

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## Conclusion

Human-wildlife conflict in India is a complex socio-ecological issue driven by habitat loss, population pressure, and climate change. Addressing it requires a **multi-pronged approach** combining **scientific management, community participation, policy**

**support, and technological innovation**. Sustainable coexistence is achievable only when both human welfare and wildlife conservation are treated as interconnected priorities.

You Asked

5. 'Cultural traditions and beliefs can serve as strong allies in environmental protection, but they can also pose risks to the health of ecosystems'. Justify the statement with appropriate examples. (10)

Kimi

Here is a comprehensive answer to the question on **Cultural Traditions and Beliefs in Environmental Protection**:

---

## Cultural Traditions and Beliefs: Allies and Risks in Environmental Protection

### Introduction

Cultural traditions and beliefs have shaped human interactions with nature for millennia. They can act as powerful tools for conservation by fostering respect for nature and regulating resource use. However, certain practices rooted in tradition can also harm ecosystems when they conflict with ecological sustainability or are practiced at unsustainable scales.

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### Part A: Cultural Traditions as Strong Allies in Environmental Protection

#### 1. Sacred Groves (Devrai / Devarakadu)

- **Description:** Forest patches protected by local communities due to religious beliefs.
- **Example:** The **sacred groves of Maharashtra and Karnataka** (e.g., Mawphlang in Meghalaya) are biodiversity-rich areas where cutting trees or hunting is strictly prohibited.
- **Conservation Value:** These groves harbor endemic species, act as gene banks, and maintain ecological balance in fragmented landscapes.

#### 2. Sacred Rivers and Water Bodies

- **Description:** Rivers like the **Ganga, Yamuna, and Narmada** are considered sacred in Hinduism.
- **Conservation Value:** This reverence has historically protected water quality and aquatic life. Many temples and rituals emphasize the purity of water, indirectly promoting water conservation.

#### 3. Animal Worship and Totemism

- **Description:** Many Indian communities worship animals (e.g., **cows, snakes, elephants, monkeys**) as deities or totems.
- **Example:** The **Ganesha** (elephant-headed god) cult promotes elephant reverence. **Nag Panchami** protects snakes.
- **Conservation Value:** Such beliefs reduce hunting and persecution of these species, aiding their survival.

#### 4. Traditional Ecological Knowledge (TEK)

- **Description:** Indigenous practices of sustainable agriculture, forestry, and fishing passed down generations.

- **Example:** The **Bishnoi community** of Rajasthan (followers of Guru Jambheshwar) strictly prohibit cutting trees and killing animals. The **Khejri** tree and **blackbuck** are sacred to them.
- **Conservation Value:** The Bishnois' protection of the Khejri tree and wildlife in the Thar Desert is a remarkable example of faith-driven conservation.

#### 5. Taboos and Rituals Regulating Resource Use

- **Description:** Many communities have seasonal taboos on fishing, hunting, or harvesting.
- **Example:** The **Tengnoupal** community in Manipur prohibits hunting during breeding seasons.
- **Conservation Value:** These taboos function as informal resource management systems, preventing overexploitation.

#### 6. Festivals Promoting Environmental Awareness

- **Example: Van Mahotsav** (tree planting festival), **Pongal** (harvest thanksgiving to nature), and **Bihu** (agricultural cycle celebration).
- **Conservation Value:** These festivals reinforce the human-nature connection and encourage sustainable practices.

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### Part B: Cultural Traditions and Beliefs Posing Risks to Ecosystem Health

#### 1. Ritualistic Pollution of Water Bodies

- **Description:** Immersion of idols, floral offerings, and cremation ashes in rivers.
- **Example:** During **Durga Puja** and **Ganesh Chaturthi**, plaster-of-Paris idols painted with toxic chemicals are immersed in rivers and seas.
- **Ecological Risk:** Heavy metals (lead, mercury), synthetic paints, and non-biodegradable materials severely pollute water, kill aquatic life, and degrade water quality.

#### 2. Mass Animal Sacrifice

- **Description:** Ritual killing of animals during religious festivals.
- **Example:** The **Gadhimai festival** in Nepal (near Indian border) and some local temples in India witness mass slaughter of goats, buffaloes, and birds.
- **Ecological Risk:** Unsanitary disposal of carcasses spreads disease, disrupts local scavenger populations, and causes public health hazards.

#### 3. Firecracker Use During Festivals

- **Description:** Extensive use of firecrackers during **Diwali**.
- **Ecological Risk:** Causes severe air pollution (PM2.5, $SO_2$, $NO_x$), noise pollution, and fire hazards. It affects human health, pets, and wildlife (especially birds and bats).

#### 4. Overexploitation of Medicinal Plants

- **Description:** Traditional medicine systems (Ayurveda, Unani, Siddha) rely on wild-harvested herbs.
- **Example:** Unsustainable collection of **Kutki** (*Picrorhiza kurroa*), **Jatamansi** (*Nardostachys jatamansi*), and **Ashwagandha** has pushed these species toward endangerment.
- **Ecological Risk:** Overharvesting without cultivation or regulation depletes wild populations and disrupts forest ecosystems.

#### 5. Religious Processions and Pilgrimages

- **Description:** Mass gatherings at religious sites (e.g., **Kumbh Mela**, **Amarnath Yatra**).

- **Ecological Risk:** Generates enormous waste, strains local water resources, causes deforestation for firewood, and tramples fragile alpine meadows.

### 6. Beliefs Leading to Invasive Species Introduction

- **Description:** Releasing animals or fish as religious merit (*punya*).
- **Example:** Release of **turtles, fish, and exotic birds** in non-native habitats.
- **Ecological Risk:** Introduced species can become invasive, outcompeting native species and disrupting local ecosystems.

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## Conclusion

Cultural traditions and beliefs are **double-edged swords** in environmental conservation. When aligned with ecological principles, they create powerful, community-driven conservation models like sacred groves and Bishnoi practices. However, when traditions ignore ecological limits — such as water pollution from idol immersion or overharvesting of medicinal plants — they become threats to ecosystem health.

The path forward lies in **adaptive management**: respecting cultural values while reforming harmful practices through education, eco-friendly alternatives (e.g., clay idols, organic colors), and integrating traditional knowledge with modern science. Sustainable coexistence requires balancing cultural identity with ecological responsibility.

You Asked

1. **Compulsory question**

(a) Match the organizations with their respective areas/ responsibilities (1×5=5)

(i) IUCN — a. Quasi-judicial body to adjudicate environmental cases

(ii) NGT — b. Framework for Sustainable development Goals

(iii) NTCA — c. maintains Red list of threatened species database

(iv) MoEFCC — d. Statutory body for tiger conservation

(v) UN — f. Monitors the National Green Corps programme

Kimi

Here are the correct matches:

| | Organization | | Area/Responsibility |
|---|---|---|---|
| **(i)** | IUCN | → | **c.** maintains Red list of threatened species database |
| **(ii)** | NGT | → | **a.** Quasi-judicial body to adjudicate environmental cases |
| **(iii)** | NTCA | → | **d.** Statutory body for tiger conservation |

| | Organization | | Area/Responsibility |
|---|---|---|---|
| **(iv)** | MoEFCC | → | **f.** Monitors the National Green Corps programme |
| **(v)** | UN | → | **b.** Framework for Sustainable development Goals |

### Brief Explanations:

- **IUCN (International Union for Conservation of Nature)** — Maintains the **Red List of Threatened Species**, the world's most comprehensive inventory of the global conservation status of biological species.
- **NGT (National Green Tribunal)** — Established in 2010 under the NGT Act; it is a **quasi-judicial body** that handles environmental disputes and cases related to environmental protection and conservation in India.
- **NTCA (National Tiger Conservation Authority)** — A **statutory body** under the Wildlife (Protection) Act, 1972, constituted in 2006 for **tiger conservation** in India. It oversees Project Tiger.
- **MoEFCC (Ministry of Environment, Forest and Climate Change)** — The nodal ministry in India for environmental protection. It **monitors the National Green Corps (NGC)** programme, which establishes eco-clubs in schools for environmental awareness.
- **UN (United Nations)** — Formulated the **2030 Agenda for Sustainable Development**, which includes the **17 Sustainable Development Goals (SDGs)** as a framework for global sustainable development.

You Asked

(a) Fill in the blanks : (1×5=5)

(i) The total number of megadiversity countries in the world is ________ .

(ii) The Wildlife (Protection) Act was first enacted in the year ________ .

(iii) Carbon footprint is the total amount of ________ gases, released by human activities.

(iv) There are ________ national missions as part of the NAPCC.

(v) ________ is an example of endemic species in India.

Kimi

Here are the answers to the fill-in-the-blanks:

**(i)** The total number of megadiversity countries in the world is **17**.

**(ii)** The Wildlife (Protection) Act was first enacted in the year **1972**.

**(iii)** Carbon footprint is the total amount of **greenhouse** gases, released by human activities.

**(iv)** There are **8** national missions as part of the NAPCC.

**(v) Lion-tailed Macaque** (or *Nilgiri Tahr / Purple Frog / Salim Ali's Fruit Bat*) is an example of endemic species in India.

---

### Brief Explanations:

- **Megadiversity countries** — The 17 countries identified by Conservation International that harbor the majority of Earth's species. India is one of them.
- **Wildlife (Protection) Act, 1972** — India's umbrella legislation for wildlife protection, which established the framework for protected areas (National Parks, Wildlife Sanctuaries) and schedules of protected species.
- **Greenhouse gases** — Include $CO_2$, $CH_4$, $N_2O$, and fluorinated gases that trap heat in the atmosphere.
- **NAPCC (National Action Plan on Climate Change)** — Launched in 2008, it comprises **8 national missions**: Solar, Enhanced Energy Efficiency, Sustainable Habitat, Water, Himalayan Ecosystem, Green India, Sustainable Agriculture, and Strategic Knowledge for Climate Change.
- **Endemic species** — Species found only in a specific geographic area. The **Lion-tailed Macaque** (*Macaca silenus*) is endemic to the Western Ghats of India.

You Asked

2. Differentiate between the following (attempt any two) (2×5=10)

(a) Ramsar Convention vs Chemical Weapons convention

(b) In-situ strategy vs ex-situ strategy for conservation

(c) Keystone species vs flagship species

Kimi

Here are the differentiations for all three pairs — you may choose any two for your answer:

---

### (a) Ramsar Convention vs Chemical Weapons Convention

| Basis | Ramsar Convention | Chemical Weapons Convention (CWC) |
|---|---|---|
| **Full Name** | Convention on Wetlands of International Importance | Convention on the Prohibition of the Development, Production, Stockpiling and Use of Chemical Weapons |
| **Year Adopted** | 1971 (signed in Ramsar, Iran) | 1993 (entered into force in 1997) |
| **Focus Area** | Conservation and sustainable use of wetlands. | Prohibition of chemical weapons and destruction of existing stockpiles. |
| **Governing Body** | Ramsar Secretariat (hosted by IUCN in Gland, Switzerland). | Organisation for the Prohibition of Chemical Weapons (OPCW) in The Hague, Netherlands. |
| **Key Objective** | To stop the loss and degradation of wetlands and promote their wise use. | To eliminate chemical weapons worldwide and prevent their re-emergence. |
| **India's Participation** | India is a party; has 85+ Ramsar sites (e.g., Keoladeo, Chilika, Sundarbans). | India ratified in 1996; declared and destroyed its chemical weapon stockpiles. |
| **Nature** | Environmental/Conservation treaty. | Disarmament/Security treaty. |

### (b) In-situ Strategy vs Ex-situ Strategy for Conservation

| Basis | In-situ Conservation | Ex-situ Conservation |
|---|---|---|
| **Definition** | Conservation of species in their natural habitats. | Conservation of species outside their natural habitats. |
| **Location** | Natural ecosystems — forests, wetlands, grasslands. | Artificial or controlled environments — zoos, botanical gardens, seed banks. |
| **Methods** | National Parks, Wildlife Sanctuaries, Biosphere Reserves, Sacred Groves. | Zoos, Aquariums, Botanical Gardens, Seed Banks, Gene Banks, Tissue Culture. |
| **Advantage** | Preserves entire ecosystems, genetic diversity, and ecological processes. | Provides intensive care, breeding programs, and protection from threats. |
| **Disadvantage** | Vulnerable to habitat destruction, poaching, and climate change. | Limited genetic diversity, artificial conditions may not suit species, expensive. |
| **Example** | Project Tiger, Project Elephant, Gir National Park. | Delhi Zoo, Indian Botanical Garden (Howrah), National Bureau of Plant Genetic Resources. |
| **Best For** | Species with large habitat requirements and stable populations. | Critically endangered species, species with very small populations, germplasm storage. |

### (c) Keystone Species vs Flagship Species

| Basis | Keystone Species | Flagship Species |
|---|---|---|
| **Definition** | A species that has a disproportionately large effect on its ecosystem relative to its abundance. | A charismatic species chosen to represent an environmental cause and raise public awarene |

| Basis | Keystone Species | Flagship Species |
|---|---|---|
| | | ss. |
| **Ecological Role** | Critical for maintaining ecosystem structure and function; their removal causes ecosystem collapse. | Primarily used for conservation marketing and fundraising; may or may not have a critical ecological role. |
| **Example** | **Sea otters** (control sea urchin populations, protect kelp forests); **Bees** (pollinators); **Wolves** (regulate prey populations). | **Royal Bengal Tiger**, **Giant Panda**, **Asian Elephant**, **Snow Leopard**. |
| **Basis of Selection** | Ecological importance and impact on community structure. | Public appeal, charisma, cultural significance, and ability to attract funding. |
| **Size/Appearance** | Can be small and inconspicuous (e.g., fig wasps, fungi, sea stars). | Usually large, charismatic, and visually appealing. |
| **Conservation Use** | Used to identify and protect critical ecological interactions. | Used as symbols for campaigns, logos, and to generate public support for conservation. |
| **Overlap** | Some species can be both (e.g., Tiger is a flagship species and also acts as an apex predator/keystone in some contexts). | Not all flagship species are keystone (e.g., Panda has limited direct ecological impact). |

You Asked

3. Write short notes on any four of the following (200 words each) (4×2.5=10)

(a) Project tiger

(b) Environment Protection Act, 1986

(c) Earth Summit

(d) Appiko movement

(e) Species reintroduction and translocation

Kimi

Here are short notes on all five topics — you may choose any four:

---

## (a) Project Tiger

**Project Tiger** is India's flagship wildlife conservation programme, launched on **April 1, 1972**, in response to the alarming decline in tiger populations due to poaching and habitat loss. It was initiated at **Jim Corbett National Park** in Uttarakhand.

**Objectives:**

- Ensure viable populations of Bengal tigers in their natural habitats.
- Preserve areas of biological importance as natural heritage.
- Maintain tiger populations for ecological, scientific, economic, aesthetic, and cultural values.

**Key Features:**

- **Tiger Reserves:** Designated protected areas with a core (critical tiger habitat) and buffer zone. India now has **54 tiger reserves** (as of 2024).
- **NTCA:** The **National Tiger Conservation Authority** was constituted in 2006 under the Wildlife (Protection) Act to oversee Project Tiger.
- **Monitoring:** Uses **M-STrIPES** (Monitoring System for Tigers — Intensive Protection and Ecological Status) for camera trap-based population estimation.

**Success:** India is home to over **75% of the world's wild tigers**, with the 2022 census recording **3,682 tigers** — the highest globally. Project Tiger is considered one of the most successful species conservation programmes worldwide.

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## (b) Environment Protection Act, 1986

The **Environment (Protection) Act, 1986** was enacted in the aftermath of the **Bhopal Gas Tragedy (1984)** to provide a comprehensive framework for environmental protection in India.

**Key Provisions:**

- It is an **umbrella legislation** that empowers the Central Government to take measures for protecting and improving the environment, including coordination with State Governments.
- The Act defines "**environment**" broadly — including water, air, land, and the interrelationship between them and human beings, other living creatures, plants, and property.
- It empowers the government to set **environmental standards**, regulate industrial emissions, and restrict areas for industrial operations.
- It provides for **penalties** (up to 5 years imprisonment and ₹1 lakh fine) for violations.

**Significance:**

- It led to the creation of the **Ministry of Environment and Forests** (now MoEFCC).
- It serves as the parent Act for various rules: **Hazardous Waste Rules, Biomedical Waste Rules, Plastic Waste Rules, E-Waste Rules, and Coastal Regulation Zone (CRZ) Notifications.**
- It remains the cornerstone of India's environmental regulatory framework.

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## (c) Earth Summit

The **Earth Summit** (officially the **UN Conference on Environment and Development — UNCED**) was held in **Rio de Janeiro, Brazil, in June 1992**. It was the largest gathering of world leaders at that time, with over 170 countries participating.

**Key Outcomes:**

1. **Rio Declaration on Environment and Development** — 27 principles guiding sustainable development, including the **polluter pays principle** and **precautionary principle**.
2. **Agenda 21** — A comprehensive blueprint for global action towards sustainable development in the 21st century.
3. **UN Framework Convention on Climate Change (UNFCCC)** — Led to the Kyoto Protocol and Paris Agreement.
4. **Convention on Biological Diversity (CBD)** — For conservation of biodiversity, sustainable use, and fair benefit-sharing.
5. **Statement of Forest Principles** — Non-legally binding principles for sustainable forest management.

**Significance for India:**

- India ratified all conventions. The Rio principles influenced India's environmental policies, including the National Environment Policy, 2006, and the NAPCC.

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## (d) Appiko Movement

The **Appiko Movement** was a grassroots environmental movement in the **Uttara Kannada district of Karnataka**, inspired by the Chipko Movement of the Himalayas. It began in **September 1983** under the leadership of **Pandurang Hegde** and local environmentalists.

**Background:**

- The movement was a response to large-scale **commercial felling of trees** by the Forest Department, which threatened the livelihoods of local communities dependent on forests for fuel, fodder, and minor forest products.

**Method:**

- Protesters **hugged trees** (*appiko* means "to hug" in Kannada) to prevent them from being cut — similar to Chipko's tree-hugging tactic.
- They also used **padayatras** (foot marches), **folk songs**, and **street plays** to spread awareness.

**Achievements:**

- Successfully halted commercial felling in several forest areas.
- Led to the adoption of **sustainable forest management practices**.
- Raised awareness about the rights of local communities over forest resources.
- Contributed to the broader discourse on **social forestry** and **joint forest management** in India.

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## (e) Species Reintroduction and Translocation

**Species Reintroduction** and **Translocation** are conservation tools used to restore or establish populations of species in areas where they have declined or become extinct.

### Species Reintroduction

- **Definition:** The deliberate release of a species into the wild in an area within its **historic range** from which it has disappeared.
- **Purpose:** To re-establish a self-sustaining wild population.
- **Example:** The **reintroduction of the Cheetah** in India (Kuno National Park, Madhya Pradesh, 2022) from Namibia and South Africa. The **Gaur** was reintroduced to Bandhavgarh from Kanha.

### Translocation

- **Definition:** The capture, transport, and release of organisms from one location to another — either within or outside their existing range.
- **Types:**
  - **Supplementation:** Adding individuals to an existing population to increase genetic diversity.
  - **Reintroduction:** Releasing into former range.
  - **Conservation Introduction:** Moving to a new area outside historic range when no suitable habitat remains in the original range.
- **Example: Asiatic Lions** from Gir to Kuno (attempted). **Rhinos** translocated between Kaziranga and Manas.

**Challenges:** Genetic bottlenecks, disease transmission, habitat suitability, human-wildlife conflict, and high costs. Both require careful planning, monitoring, and stakeholder involvement.

You Asked

4. (a) With appropriate reasons, discuss why India is considered as a megadiversity nation. (5)

(b) Critically examine the importance of sacred groves in biodiversity conservation and protection of indigenous cultures. (5)

Kimi

Here are comprehensive answers to both parts:

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## 4. (a) Why India is Considered a Megadiversity Nation

India is recognized as one of the **17 megadiversity countries** of the world, a designation given by Conservation International to nations that harbor the majority of Earth's species. Several factors contribute to this status:

### 1. Exceptional Species Richness

- India harbors approximately **7-8% of all recorded species** globally, despite covering only **2.4% of the world's land area**.
- It has over **91,000 animal species** and **45,000 plant species**, with high levels of endemism.

### 2. Diverse Biogeographic Zones

- India spans **10 biogeographic zones**: Trans-Himalayan, Himalayan, Desert, Semi-Arid, Western Ghats, Deccan Peninsula, Gangetic Plain, North-East, Coast, and Islands.
- Each zone supports distinct ecosystems and species assemblages.

### 3. High Endemism

- The **Western Ghats** and **Eastern Himalayas** are biodiversity hotspots with numerous endemic species.
- Examples: **Lion-tailed Macaque, Nilgiri Tahr, Purple Frog, Asiatic Lion** (found only in Gir).

#### 4. Varied Ecosystems

- India possesses a wide range of ecosystems: tropical rainforests, alpine meadows, mangroves, coral reefs, deserts, wetlands, and grasslands.
- Major ecosystems include the **Sundarbans** (largest mangrove forest), **Western Ghats** (UNESCO World Heritage Site), and the **Himalayan range**.

#### 5. Two Major Biodiversity Hotspots

- India contains parts of **two of the world's 36 biodiversity hotspots**: the **Himalayas** and the **Western Ghats-Sri Lanka** hotspot.
- Hotspots are regions with exceptional species richness and high degree of habitat loss.

#### 6. Rich Agrobiodiversity

- India is a **centre of origin** for several crop species (rice, sugarcane, mango, turmeric).
- It maintains vast genetic diversity through traditional farming and indigenous varieties.

#### 7. Marine and Coastal Biodiversity

- India's **7,500 km coastline** supports coral reefs, seagrass beds, and rich marine fisheries.
- The **Andaman & Nicobar** and **Lakshadweep** islands add unique insular biodiversity.

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## 4. (b) Importance of Sacred Groves in Biodiversity Conservation and Protection of Indigenous Cultures

### Introduction

Sacred groves are **forest patches protected by local communities** due to religious beliefs, cultural traditions, or spiritual associations. Known as **Devrai** (Maharashtra), **Devarakadu** (Karnataka), **Oran** (Rajasthan), and **Kavu** (Kerala), these groves represent one of the oldest forms of community-based conservation in India.

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### Importance in Biodiversity Conservation

#### 1. Refugia for Endemic and Threatened Species

- Sacred groves often harbor **rare, endemic, and threatened species** that have disappeared from surrounding landscapes due to deforestation.
- Example: The **Mawphlang Sacred Grove** in Meghalaya protects ancient trees, orchids, and medicinal plants not found elsewhere in the region.

#### 2. Gene Pools and Genetic Diversity

- These groves act as **in-situ gene banks**, preserving genetic diversity of wild relatives of crops and medicinal plants.
- They maintain **old-growth forests** with complex ecological structures.

#### 3. Ecosystem Services

- **Water conservation:** Groves protect watersheds and maintain groundwater recharge.
- **Soil protection:** Prevent erosion and maintain soil fertility in surrounding agricultural lands.
- **Microclimate regulation:** Provide local cooling effects and maintain humidity.

#### 4. Wildlife Habitat

- Serve as stepping stones and corridors for wildlife movement in fragmented landscapes.
- Protect breeding and foraging grounds for birds, insects, and small mammals.

#### 5. Medicinal Plant Reservoirs

- Many sacred groves contain **Ayurvedic and traditional medicinal plants** that are overharvested elsewhere.
- Example: The **Khasi sacred groves** protect numerous medicinal orchids and ferns.

---

### Importance in Protection of Indigenous Cultures

#### 1. Living Archives of Traditional Knowledge

- Sacred groves embody **Traditional Ecological Knowledge (TEK)** passed down through generations.
- They preserve knowledge of plant uses, seasonal cycles, and sustainable harvesting practices.

#### 2. Religious and Spiritual Significance

- Groves are associated with **local deities, spirits, and ancestors**.
- Rituals and festivals tied to groves reinforce community bonds and cultural identity.
- Example: The **Bishnoi community** protects the **Khejri** tree as sacred, reflecting deep environmental ethics.

#### 3. Community Governance and Social Cohesion

- Management of sacred groves follows **community rules and taboos**, strengthening local governance.
- Collective responsibility for grove protection fosters social unity and intergenerational learning.

#### 4. Cultural Landscapes

- Sacred groves are part of **cultural landscapes** that link people to their land.
- They represent the **biocultural heritage** of indigenous and rural communities.

---

### Critical Examination: Challenges and Limitations

| Challenge | Explanation |
|---|---|
| **Erosion of Beliefs** | Modernization and urbanization weaken traditional beliefs, reducing community protection. |
| **Fragmentation** | Many groves are small and isolated, limiting their ecological effectiveness. |
| **Lack of Legal Recognition** | Most groves are not formally protected under wildlife laws, making them vulnerable to encro achment. |
| **Population Pressure** | Growing human populations increase demands on land, threatening grove boundaries. |
| **Invasive Species** | Some groves are invaded by exotic species like *Lantana* and *Parthenium*. |

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### Conclusion

Sacred groves represent a **unique intersection of ecology and culture**. They are vital for conserving biodiversity in fragmented landscapes and preserving the biocultural heritage of India's indigenous communities. Integrating sacred grove conservation into formal protected area networks while respecting community rights is essential for their long-term survival.

You Asked

5. Discuss the challenges and policy measures related to the resettlement and rehabilitation of people and communities displaced by developmental projects in India. (10)

Kimi

Here is a comprehensive answer to the question on **Resettlement and Rehabilitation of Displaced Communities in India**:

---

## Resettlement and Rehabilitation of People Displaced by Developmental Projects in India

### Introduction

Developmental projects in India — including dams, mines, highways, industrial corridors, and urban expansion — have displaced millions of people, particularly **tribal communities, farmers, and marginalized groups**. Resettlement and Rehabilitation (R&R) remains one of the most contentious socio-environmental issues, balancing national development with social justice and human rights.

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### Part A: Challenges in Resettlement and Rehabilitation

#### 1. Inadequate Compensation and Delayed Payments

- Displaced families often receive **below-market compensation** for land and assets.
- Delayed disbursement of compensation creates financial distress and forces distress migration.
- Many lack proper land titles (especially tribals with customary rights), making them ineligible for compensation.

#### 2. Loss of Livelihoods and Economic Disruption

- Agricultural communities lose not just land but also **access to forests, water bodies, and common property resources**.
- Skills mismatch in new locations — farmers struggle to find non-farm employment.
- **Wage laborers** dependent on local agricultural economies are often ignored in R&R frameworks.

#### 3. Social and Cultural Disintegration

- Displacement breaks **community bonds, kinship networks, and social support systems**.
- Tribal communities lose connection to **sacred sites, ancestral lands, and cultural identity**.
- Women face disproportionate burdens — loss of access to fuelwood, water, and traditional roles.

#### 4. Poor Quality of Resettlement Sites

- Resettlement colonies often lack **basic infrastructure**: drinking water, sanitation, electricity, schools, and healthcare.
- Sites are frequently located on **marginal, infertile lands** far from original habitats.
- **Sardar Sarovar Dam** displaced thousands to poorly planned resettlement sites in Gujarat and Maharashtra.

#### 5. Lack of Informed Consent and Participation

- Many projects proceed with **minimal consultation** with affected communities.
- The principle of **Free, Prior, and Informed Consent (FPIC)** is often violated, especially for tribal communities under **Schedule V and VI areas**.
- **Narmada Bachao Andolan** highlighted how affected communities were not genuinely consulted.

#### 6. Legal and Administrative Hurdles

- Complex bureaucratic procedures delay R&R implementation.
- Weak enforcement of existing laws; project proponents often prioritize construction over rehabilitation.
- **Land acquisition** processes sometimes use **eminent domain** coercively.

#### 7. Psychological Trauma and Health Impacts

- Displacement causes **stress, anxiety, and trauma** ("development-induced displacement stress").
- Loss of traditional food sources and healthcare knowledge affects nutritional and health outcomes.
- Increased vulnerability to **alcoholism and substance abuse** in resettled communities.

#### 8. Gender Inequity

- Women are often excluded from compensation negotiations and land titles.
- Loss of traditional livelihoods (gathering, livestock rearing) disproportionately affects women.
- Increased domestic violence and reduced decision-making power in new settlements.

---

### Part B: Policy Measures and Legal Frameworks

#### 1. The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (RFCTLARR Act)

- Replaced the colonial-era Land Acquisition Act of 1894.
- **Key provisions:**
  - **Social Impact Assessment (SIA)** mandatory for private and public-private partnership projects.
  - **Informed consent** required from 80% of affected families for private projects and 70% for PPP projects.
  - **Compensation** at up to **4 times market value** in rural areas and **2 times** in urban areas.
  - **Rehabilitation and Resettlement** entitlements for all affected families, including landless laborers.
  - **Resettlement allowance, subsistence allowance, and employment** provisions.

#### 2. National Rehabilitation and Resettlement Policy, 2007

- Predecessor to the 2013 Act; provided the first comprehensive national framework.
- Emphasized **minimum displacement**, **improvement in living standards**, and **community participation**.
- Required **resettlement at least 6 months before** displacement.

#### 3. Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA)

- Recognizes **individual and community forest rights** of tribal and forest-dwelling communities.
- Requires **gram sabha consent** for displacement and resettlement.
- Protects tribal rights over land and resources, reducing arbitrary evictions.

#### 4. Panchayats (Extension to Scheduled Areas) Act, 1996 (PESA)

- Grants **autonomy to gram sabhas** in Schedule V areas to approve or reject land acquisition and projects.
- Mandates **consultation with tribal communities** before any displacement.

#### 5. National Policy on Tribals, 2006

- Emphasizes **tribal-centric development** and protection of tribal land and culture.
- Advocates for **alternative livelihoods** and **capacity building** in resettlement.

#### 6. Supreme Court and NGT Interventions

- The Supreme Court has issued guidelines for **fair compensation, rehabilitation before displacement, and livelihood restoration**.
- **National Green Tribunal (NGT)** addresses environmental and displacement-related grievances.

#### 7. Special Packages for Vulnerable Groups

- **Special Component Plans** for Scheduled Castes and Scheduled Tribes.
- **Women-headed household** provisions under the 2013 Act.
- **Skill development and vocational training** programmes for resettled youth.

---

### Part C: Persistent Gaps and Recommendations

| Gap | Recommendation |
|---|---|
| Weak implementation of the 2013 Act | Strengthen monitoring mechanisms and penalties for non-compliance. |
| Exclusion of urban displaced | Extend R&R coverage to urban redevelopment and slum rehabilitation. |
| Lack of livelihood restoration | Mandate **employment guarantees** and **skill training** in resettlement plans. |
| Cultural insensitivity | Involve anthropologists and social workers in planning culturally appropriate resett lement. |
| No tracking of long-term outcome s | Establish **long-term monitoring** of resettled communities' well-being. |

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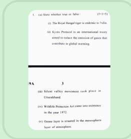
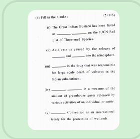
### Conclusion

Resettlement and rehabilitation in India remain fraught with challenges of **equity, participation, and implementation**. While progressive legal frameworks like the **RFCTLARR Act, 2013** and **FRA, 2006** provide a strong foundation, ground-level realities often fall short. Genuine rehabilitation requires **participatory planning, timely compensation, livelihood restoration, and respect for cultural identity**. Development must not come at the cost of the most vulnerable — sustainable progress demands that no community is left behind.

You Asked

Kimi

Here are the answers to both sections:

---

### 1. (a) True or False

| | Statement | Answer | Explanation |
|---|---|---|---|
| (i) | The Royal Bengal tiger is endemic to India. | **False** | The Royal Bengal tiger (*Panthera tigris tigris*) is found in India, Bangladesh, Nepal, Bhutan, and Myanmar. It is **not endemic** to India alone. |
| (ii) | Kyoto Protocol is an international treaty aimed to reduce the emission of gases that contribute to global warming. | **True** | The Kyoto Protocol (1997) commits industrialized countries to reduce greenhouse gas emissions. |
| (iii) | Silent Valley movement took place in Uttarakhand. | **False** | The Silent Valley movement took place in **Kerala** (Western Ghats), not Uttarakhand. |
| (iv) | Wildlife Protection Act came into existence in the year 1972. | **True** | The Wildlife (Protection) Act was enacted in **1972** and came into force in 1972. |
| (v) | Ozone layer is situated in the mesosphere layer of atmosphere. | **False** | The ozone layer is situated in the **stratosphere** (approximately 10–50 km above Earth's surface), not the mesosphere. |

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### 1. (b) Fill in the Blanks

| | Answer |
|---|---|
| (i) | The Great Indian Bustard has been listed as **Critically Endangered** on the IUCN Red List of Threatened Species. |
| (ii) | Acid rain is caused by the release of **Sulphur dioxide ($SO_2$)** and **Nitrogen oxides ($NO_x$)** into the atmosphere. |
| (iii) | **Diclofenac** is the drug that was responsible for large scale death of vultures in the Indian subcontinent. |
| (iv) | **Carbon footprint** is a measure of the amount of greenhouse gases released by various activities of an individual or entity. |
| (v) | **Ramsar** Convention is an international treaty for the protection of wetlands. |

---

#### Brief Explanations:

- **Great Indian Bustard** — One of the heaviest flying birds; its population has declined drastically due to habitat loss, power line collisions, and hunting. Listed as **Critically Endangered** by IUCN.
- **Acid rain** — $SO_2$ and $NO_x$ react with water, oxygen, and other chemicals in the atmosphere to form sulfuric and nitric acids, which fall as acid precipitation.
- **Diclofenac** — A veterinary non-steroidal anti-inflammatory drug (NSAID) used in cattle. Vultures feeding on carcasses of treated animals suffered renal failure, causing a **99.9% population collapse** of three vulture species in India.
- **Carbon footprint** — Measured in tonnes of $CO_2$ equivalent ($tCO_2e$), it quantifies the total GHG emissions associated with an activity, product, or entity.
- **Ramsar Convention** — Adopted in 1971 in Ramsar, Iran; provides the framework for national action and international cooperation for the conservation and wise use of wetlands.

You Asked

2. Differentiate between the following (attempt any two) (2×5=10)

(a) National Green Tribunal and National Green Corps

(b) Biosphere Reserve and Biodiversity Hotspot

(c) In-situ conservation and ex-situ conservation

Kimi

Here are the differentiations for all three pairs — you may choose any two:

---

### (a) National Green Tribunal (NGT) vs National Green Corps (NGC)

| Basis | National Green Tribunal (NGT) | National Green Corps (NGC) |
|---|---|---|
| Nature | A **quasi-judicial body** established under the NGT Act, 2010. | An **environmental education programme** for school students, launched by MoEFCC. |
| Purpose | To adjudicate environmental disputes and enforce environmental laws. | To create environmental awareness among students and promote conservation activities. |
| Composition | Comprises judicial members (retired judges) and expert members (environmental scientists). | Comprises **Eco-clubs** formed in schools with student members and teacher coordinators. |
| Functions | Hears cases related to environmental protection, forest conservation, and pollution control. | Organizes tree plantations, cleanliness drives, waste management, and awareness campaigns. |
| Legal Authority | Can pass binding orders, impose fines, and direct compensation. | Has no legal authority; functions as an educational and voluntary initiative. |

| Basis | National Green Tribunal (NGT) | National Green Corps (NGC) |
|---|---|---|
| **Headquarters** | Principal Bench in New Delhi; regional benches in Pune, Chennai, Kolkata, Bhopal. | Operates through schools across India under the guidance of State nodal agencies. |
| **Example of Work** | Banned diesel vehicles older than 10 years in Delhi-NCR; ordered compensation for industrial pollution. | Millions of eco-clubs participate in World Environment Day, Van Mahotsav, and local conservation projects. |

---

### (b) Biosphere Reserve vs Biodiversity Hotspot

| Basis | Biosphere Reserve | Biodiversity Hotspot |
|---|---|---|
| **Definition** | A designated protected area recognized by UNESCO's Man and the Biosphere (MAB) Programme for conservation and sustainable development. | A biogeographic region with exceptional species richness, high endemism, and significant habitat loss (at least 70%). |
| **Governing Body** | UNESCO (United Nations Educational, Scientific and Cultural Organization). | Conservation International (CI) originally identified the concept; no formal governing body. |
| **Zones** | Divided into three zones: **Core** (strict protection), **Buffer** (limited activity), and **Transition** (sustainable development). | No zonation; it is a conceptual designation based on biological criteria. |
| **Focus** | Conservation + research + sustainable development + community involvement. | Conservation prioritization based on irreplaceability and vulnerability. |
| **Criteria for Selection** | Ecological significance, potential for research, and capacity for sustainable development. | Must have **≥1,500 endemic plant species** and have lost **≥70% of original habitat**. |
| **Number** | India has **18 biosphere reserves** (e.g., Nilgiri, Sundarbans, Nanda Devi, Pachmarhi). | There are **36 global hotspots**; India contains parts of **two**: the **Himalayas** and the **Western Ghats-Sri Lanka**. |
| **Human Activity** | Encourages sustainable human activity in transition zones. | Human activity is generally the cause of habitat loss that qualifies an area as a hotspot. |

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### (c) In-situ Conservation vs Ex-situ Conservation

| Basis | In-situ Conservation | Ex-situ Conservation |
|---|---|---|
| **Definition** | Conservation of species in their **natural habitats**. | Conservation of species **outside their natural habitats** in artificial or controlled environments. |
| **Location** | Natural ecosystems — forests, wetlands, grasslands, oceans. | Zoos, botanical gardens, aquariums, seed banks, gene banks, tissue culture labs. |
| **Methods** | National Parks, Wildlife Sanctuaries, Biosphere Reserves, Sacred Groves, Community Reserves. | Zoos, Aquariums, Botanical Gardens, Seed Banks, Cryopreservation, Captive Breeding Centres. |

| Basis | In-situ Conservation | Ex-situ Conservation |
|---|---|---|
| **Advantages** | Preserves entire ecosystems, natural evolutionary processes, genetic diversity, and species interactions. | Provides intensive care, protection from immediate threats, controlled breeding, and research opportunities. |
| **Disadvantages** | Vulnerable to habitat destruction, poaching, climate change, and limited control over threats. | Limited genetic diversity, risk of inbreeding, animals may lose natural behaviors, expensive to maintain. |
| **Examples in India** | Project Tiger, Project Elephant, Gir National Park, Kaziranga National Park. | Delhi Zoo, Indian Botanical Garden (Howrah), National Bureau of Plant Genetic Resources (NBPGR), Frozen Zoo. |
| **Best Suited For** | Species with viable populations and intact habitats. | Critically endangered species, species with extremely small populations, germplasm storage for future use. |
| **Cost** | Relatively lower long-term cost; requires land protection. | High operational and infrastructure costs. |

You Asked

2. Differentiate between the following (**attempt any two**) (2×5=10)

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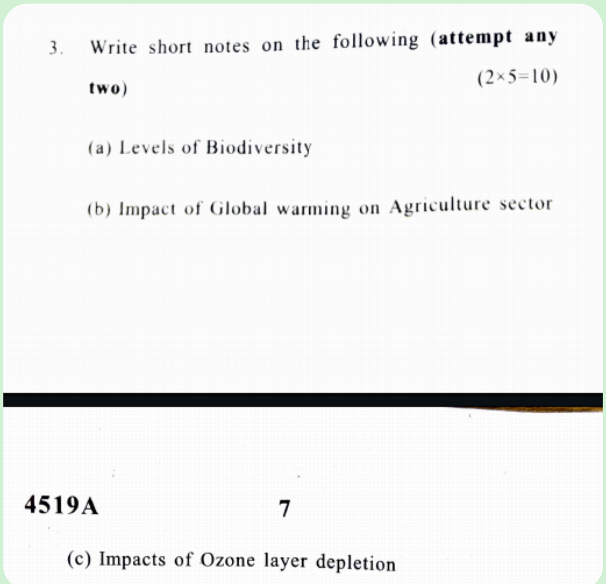
You Asked


3. Write short notes on the following (attempt any two) (2×5=10)

(a) Levels of Biodiversity

(b) Impact of Global warming on Agriculture sector

4519A 7

(c) Impacts of Ozone layer depletion

Kimi

Here are short notes on all three topics — you may choose any two:

## (a) Levels of Biodiversity

Biodiversity is the variety of life on Earth, and it is studied at three hierarchical levels:

### 1. Genetic Diversity

- **Definition:** Variation in genes within a single species.
- **Significance:** Higher genetic diversity increases a species' ability to adapt to environmental changes, resist diseases, and survive catastrophes.
- **Example:** Different varieties of rice (*Oryza sativa*) — Basmati, Sona Masuri, IR-8 — show genetic diversity within the same species.

### 2. Species Diversity

- **Definition:** Variety of different species within a particular area or ecosystem.
- **Significance:** Greater species diversity leads to more stable and resilient ecosystems. It ensures functional redundancy — if one species fails, another can perform its ecological role.
- **Example:** A tropical rainforest contains thousands of species of trees, insects, birds, and mammals, whereas a desert has far fewer species.

### 3. Ecosystem Diversity

- **Definition:** Variety of ecosystems in a given region — forests, grasslands, wetlands, deserts, coral reefs, etc.
- **Significance:** Different ecosystems provide different services (carbon sequestration, water purification, pollination). Ecosystem diversity ensures overall planetary stability.
- **Example:** India has deserts (Thar), alpine meadows (Himalayas), mangroves (Sundarbans), and coral reefs (Lakshadweep).

### Interconnection

These levels are interdependent. Loss of ecosystem diversity leads to species extinction, which reduces genetic diversity. Conservation must address all three levels for holistic biodiversity protection.

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## (b) Impact of Global Warming on Agriculture Sector

Global warming — the long-term increase in Earth's average temperature due to greenhouse gas emissions — poses severe threats to agriculture:

### 1. Altered Crop Yields

- **Temperature rise** beyond optimal levels reduces photosynthesis and grain filling in crops like wheat and rice.
- **Heat stress** during flowering causes sterility in crops like maize and sorghum.
- Studies project **10-40% yield reduction** in Indian wheat by 2100 under high-emission scenarios.

### 2. Changes in Growing Seasons

- Shifts in **phenology** — earlier flowering, fruiting, and maturity — disrupt traditional farming calendars.
- **Rabi crops** (winter crops) may face shortened growing periods due to early warming.

### 3. Water Scarcity and Irrigation Stress

- Higher temperatures increase **evapotranspiration**, raising water demand for irrigation.

- **Glacial retreat** in the Himalayas threatens river flows (Ganga, Indus, Brahmaputra) that sustain agriculture in North India.
- **Groundwater depletion** is accelerated in water-stressed regions.

### 4. Increased Pest and Disease Pressure

- Warmer climates expand the geographic range of **pests** (e.g., aphids, locusts) and **plant diseases**.
- Higher humidity and temperature favor fungal and bacterial pathogens.

### 5. Extreme Weather Events

- **Droughts, floods, and unseasonal rainfall** damage standing crops and reduce soil fertility.
- Example: The 2015 unseasonal rains destroyed millions of hectares of wheat and mustard in North India.

### 6. Soil Degradation

- Increased temperatures accelerate **soil organic matter decomposition**, reducing fertility.
- **Salinization** of coastal agricultural lands due to sea-level rise affects deltas like the Ganga-Brahmaputra.

### 7. Impact on Livestock

- Heat stress reduces milk production and fertility in dairy animals.
- Changes in pasture quality affect grazing animals.

### Adaptation Strategies

- Development of **heat-tolerant crop varieties** (e.g., climate-resilient rice).
- **Precision agriculture**, drip irrigation, and water harvesting.
- **Crop diversification** and shifting to climate-resilient crops (millets, sorghum).

---

## (c) Impacts of Ozone Layer Depletion

The ozone layer in the stratosphere absorbs harmful **UV-B radiation (280–315 nm)**. Its depletion has serious consequences:

### 1. Human Health Impacts

- **Skin Cancer:** Increased UV-B exposure raises incidence of **melanoma** and **non-melanoma skin cancers**.
- **Cataracts:** UV radiation damages the eye lens, causing cataracts and blindness.
- **Immune System Suppression:** UV-B weakens the immune system, reducing the body's ability to fight infections and increasing susceptibility to diseases.

### 2. Agricultural Impacts

- UV-B damages **plant DNA, photosynthetic apparatus, and growth hormones**.
- Sensitive crops like **soybean, wheat, and rice** show reduced yields.
- Increased UV-B alters **plant-insect interactions**, affecting pollination and pest resistance.

### 3. Marine Ecosystem Impacts

- **Phytoplankton** — the base of marine food webs — are highly sensitive to UV-B radiation.
- Reduced phytoplankton productivity affects **fish populations** and entire marine ecosystems.
- UV-B harms **fish larvae, shrimp, and crab** development stages.

#### 4. Damage to Materials

- UV-B accelerates **degradation of plastics, rubber, wood, and fabrics**.
- Increased maintenance costs for outdoor infrastructure and equipment.

#### 5. Air Quality Degradation

- UV-B increases **ground-level ozone** (a pollutant) through photochemical reactions.
- This worsens **respiratory problems** like asthma and bronchitis.

#### 6. Impact on Biogeochemical Cycles

- UV-B affects **nitrogen and sulfur cycles**, altering nutrient availability in ecosystems.
- Changes in **decomposition rates** of organic matter in soils and water bodies.

### International Response

The **Montreal Protocol (1987)** successfully phased out ozone-depleting substances (CFCs, halons). The ozone layer is slowly recovering, projected to return to 1980 levels by mid-century. This is considered the most successful environmental treaty in history.

You Asked

4. (a) India is one of the seventeen megadiversity nations globally. Substantiate this statement giving appropriate reasons and examples.

(b) What are the criteria for any region to be declared as a Biodiversity hotspot? (5+5)

Kimi

Here are comprehensive answers to both parts:

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### 4. (a) India as One of the Seventeen Megadiversity Nations

India is recognized as one of the **17 megadiversity countries** of the world, a designation given by **Conservation International** to nations that collectively harbor approximately **70% of the world's biodiversity**. Several compelling reasons substantiate this status:

#### 1. Exceptional Species Richness

- India harbors approximately **7–8% of all recorded species** globally, despite covering only **2.4% of the world's land area**.
- It has over **91,000 animal species** and **45,000 plant species**, making it one of the most species-rich nations.

#### 2. Two Global Biodiversity Hotspots

- India contains parts of **two of the world's 36 biodiversity hotspots**:
  - **The Himalayas** — Rich in endemic alpine flora and fauna.
  - **The Western Ghats-Sri Lanka hotspot** — One of the hottest hotspots with exceptional endemism.
- These hotspots are regions with irreplaceable biodiversity under severe threat.

### 3. High Endemism

- Numerous species are found **nowhere else on Earth**:
  - **Lion-tailed Macaque** (*Macaca silenus*) — Western Ghats
  - **Nilgiri Tahr** (*Nilgiritragus hylocrius*) — Western Ghats
  - **Asiatic Lion** (*Panthera leo persica*) — Gir Forest, Gujarat
  - **Purple Frog** (*Nasikabatrachus sahyadrensis*) — Western Ghats
  - **Great Indian Bustard** (*Ardeotis nigriceps*) — Indian subcontinent

### 4. Diverse Biogeographic Zones

- India spans **10 distinct biogeographic zones**: Trans-Himalayan, Himalayan, Desert, Semi-Arid, Western Ghats, Deccan Peninsula, Gangetic Plain, North-East, Coast, and Islands.
- Each zone supports unique ecosystems and species assemblages.

### 5. Varied Ecosystems

- **Tropical rainforests** (Western Ghats, North-East)
- **Alpine meadows and glaciers** (Himalayas)
- **Mangroves** (Sundarbans — world's largest)
- **Coral reefs** (Gulf of Mannar, Lakshadweep, Andaman & Nicobar)
- **Deserts** (Thar)
- **Wetlands** (Keoladeo, Chilika, Loktak)
- **Grasslands** (Terai, Deccan)

### 6. Rich Agrobiodiversity

- India is a **Vavilov centre of origin** for several crops: rice, sugarcane, mango, turmeric, black pepper, and cotton.
- It maintains vast **genetic diversity** through traditional farming and indigenous landraces.

### 7. Marine and Coastal Biodiversity

- India's **7,500 km coastline** and island territories support rich marine life.
- The **Andaman & Nicobar Islands** and **Lakshadweep** add unique insular biodiversity with high endemism.

### 8. Significant Forest Cover

- India has approximately **24% forest cover**, including tropical, subtropical, temperate, and alpine forests.
- It hosts **54 tiger reserves**, **500+ wildlife sanctuaries**, and **100+ national parks**.

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## 4. (b) Criteria for a Region to be Declared a Biodiversity Hotspot

The concept of biodiversity hotspots was introduced by **Norman Myers** in 1988 and is now managed by **Conservation International**. A region must meet **two strict criteria** to qualify:

### Criterion 1: Species Endemism

- The region must contain at least **1,500 species of vascular plants** (flowering plants, conifers, ferns) that are **endemic** (found nowhere else on Earth).
- This criterion ensures the region harbors **unique, irreplaceable biodiversity** that cannot be conserved elsewhere.

### Criterion 2: Degree of Habitat Loss

- The region must have lost at least **70% of its original primary vegetation** due to human activities.
- This criterion prioritizes regions under **severe threat**, where conservation action is most urgent.

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### Additional Characteristics (Not Formal Criteria but Important)

| Characteristic | Explanation |
|---|---|
| **High Species Richness** | Hotspots typically contain extraordinary concentrations of species across taxa. |
| **Threat from Human Activities** | Agriculture, deforestation, urbanization, mining, and climate change drive habitat loss. |
| **Ecological Significance** | Hotspots often contain unique ecosystems with specialized adaptations. |
| **Cultural and Economic Value** | Many hotspots are home to indigenous communities and provide ecosystem services. |

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### Global Hotspots and India's Share

- There are **36 recognized biodiversity hotspots** worldwide.
- India contains parts of **two hotspots**:
  1. **The Himalayas** — Covers the Himalayan range across India, Nepal, Bhutan, and Tibet; rich in endemic alpine flora.
  2. **The Western Ghats and Sri Lanka** — The Western Ghats stretch along India's west coast; together with Sri Lanka, this hotspot has exceptional plant and amphibian endemism.

### Significance of Hotspot Conservation

- Hotspots cover only **2.4% of Earth's land surface** but contain over **50% of the world's plant species** and **42% of all terrestrial vertebrate species** as endemics.
- Conservation in hotspots is **cost-effective** because it protects the maximum number of unique species per unit area.

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### Conclusion

India's megadiversity status is well-substantiated by its extraordinary species richness, high endemism, diverse ecosystems, and inclusion in two global biodiversity hotspots. The hotspot criteria of **endemism** and **habitat loss** guide global conservation priorities, ensuring that limited resources are directed toward protecting the most irreplaceable and threatened biodiversity on Earth.

You Asked

5. Human population explosion is causing severe resource depletion and environmental degradation. Justify the statement, giving appropriate examples. (10)

Kimi

Here is a comprehensive answer to the question on **Human Population Explosion and Environmental Degradation**:

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## Human Population Explosion: Causes of Resource Depletion and Environmental Degradation

### Introduction

Human population has grown exponentially from **1 billion in 1800 to over 8 billion today**. This **population explosion** — characterized by rapid growth due to declining mortality rates and high fertility in many regions — places unsustainable pressure on Earth's finite resources. The relationship is direct: more people demand more food, water, energy, and land, while generating more waste and pollution, leading to severe **resource depletion** and **environmental degradation**.

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### Part A: Resource Depletion

#### 1. Land Resource Depletion

- Growing populations require land for housing, agriculture, industry, and infrastructure.
- **Urban sprawl** consumes fertile agricultural land and forests.
- **Example:** The National Capital Region (NCR) of India has expanded rapidly, converting thousands of hectares of farmland into concrete jungles. Globally, the **Aral Sea** shrank drastically because rivers were diverted for cotton irrigation to feed a growing Soviet population.

#### 2. Water Scarcity

- Freshwater is finite; per capita availability declines as population rises.
- Over-extraction of groundwater for agriculture and urban use lowers water tables.
- **Example:** India is the world's largest groundwater user. States like **Punjab, Haryana, and Rajasthan** face critical groundwater depletion. The **Cape Town water crisis (2018)** was driven by population growth and climate stress.

#### 3. Deforestation

- Expanding populations need timber, fuelwood, and agricultural land.
- **Example:** The **Amazon rainforest** has lost over 17% of its cover in the last 50 years, largely due to cattle ranching and soy farming to meet global food demand. In India, **Jharkhand and Chhattisgarh** have experienced severe forest loss due to mining and agriculture.

#### 4. Fossil Fuel Depletion

- Industrialization and rising living standards increase energy consumption.
- **Example:** Global oil consumption has risen from **3 billion tonnes (1965) to over 4.5 billion tonnes (2023)**. India's coal reserves are depleting rapidly to meet the energy needs of 1.4 billion people.

#### 5. Mineral and Soil Depletion

- Intensive agriculture to feed growing populations strips soil nutrients.
- **Example:** The **Dust Bowl of 1930s USA** was caused by over-plowing marginal lands to meet food demand. In India, **micronutrient deficiency in soils** is widespread due to continuous cropping without replenishment.

#### 6. Overfishing

- Growing demand for protein depletes marine fish stocks.
- **Example: 90% of global fish stocks** are fully exploited or overfished. The **Indian Ocean yellowfin tuna** population has declined sharply due to overfishing.

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### Part B: Environmental Degradation

#### 1. Air Pollution

- More people = more vehicles, industries, and energy use.
- **Example: Delhi's air quality** is among the world's worst, driven by vehicular emissions, industrial pollution, and construction dust from a population of over 30 million in the NCR. Globally, **7 million premature deaths** occur annually due to air pollution (WHO).

#### 2. Water Pollution

- Untreated sewage and industrial waste from dense populations contaminate rivers and lakes.
- **Example:** The **Ganga River** receives over **1.2 billion litres of untreated sewage daily** from cities along its banks. The **Citarum River in Indonesia** is one of the world's most polluted rivers due to industrial and domestic waste from a dense population.

#### 3. Soil Degradation

- Overuse of chemical fertilizers and pesticides to boost yields degrades soil health.
- **Example: Punjab's "cancer belt"** is linked to excessive pesticide use in intensive agriculture. **Salinization** in the Indo-Gangetic plain is rising due to unsustainable irrigation.

#### 4. Loss of Biodiversity

- Habitat destruction, pollution, and overexploitation drive species extinction.
- **Example:** The **Great Indian Bustard** is critically endangered due to habitat loss from agriculture and power lines. Globally, the current extinction rate is **100–1,000 times** the natural background rate, driven primarily by human activity.

#### 5. Climate Change

- Population growth increases greenhouse gas emissions through energy use, deforestation, and agriculture.
- **Example:** India is the **3rd largest emitter** of $CO_2$ globally, largely due to coal-based energy for its 1.4 billion population. The **IPCC** confirms that anthropogenic emissions are the primary driver of global warming.

#### 6. Waste Generation

- Urban populations generate enormous solid and plastic waste.
- **Example:** India generates approximately **62 million tonnes of waste annually**, much of it unsegregated and dumped in landfills. The **Ghazipur landfill in Delhi** is a towering environmental hazard.

#### 7. Urban Heat Islands and Reduced Green Cover

- Concrete surfaces and reduced vegetation in overcrowded cities raise local temperatures.
- **Example: Mumbai, Delhi, and Kolkata** experience urban heat island effects 3–5°C higher than surrounding rural areas.

### The Vicious Cycle

Population growth → Increased resource demand → Overexploitation → Resource depletion → Environmental degradation → Reduced carrying capacity → Poverty and conflict → Further pressure on remaining resources.

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### Conclusion

The statement is thoroughly justified. Human population explosion is the underlying driver of most contemporary environmental crises. While technological solutions and better governance can mitigate impacts, **sustainable development** requires addressing both **population stabilization** (through education, family planning, and women's empowerment) and **consumption patterns** (through circular economy, renewable energy, and sustainable agriculture). Without this dual approach, resource depletion and environmental degradation will accelerate, threatening the very life-support systems humanity depends upon.

You Asked

1. (a) State whether true or false : (5×1=5)

(i) Forest Conservation Act came into existence in the year 1960.

(ii) Cryopreservation is an example of ex-situ conservation.

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(iii) India has been divided into 5 recognizable biogeographic zones.

(iv) Asiatic Lion is an example of invasive species in India.

(v) The National Green Tribunal (NGT) was set up under the NGT Act in the year 2010.

Kimi

Here are the answers to the True/False questions:

| | Statement | Answer | Explanation |
|---|---|---|---|
| **(i)** | Forest Conservation Act came into existence in the year 1960. | **False** | The Forest (Conservation) Act was enacted in **1980**, not 1960. It was amended in 1988. |
| **(ii)** | Cryopreservation is an example of ex-situ conservation. | **True** | Cryopreservation (preservation of cells, tissues, or embryos at ultra-low temperatures) is a technique used in **ex-situ conservation** to maintain genetic material outside natural habitats. |
| **(iii)** | India has been divided into 5 recognizable biogeographic zones. | **False** | India has been divided into **10 biogeographic zones**, not 5. These are: Trans-Himalayan, Himalayan, Desert, Semi-Arid, Western Ghats, Deccan Peninsula, Gangetic Plain, North-East, Coast, and Islands. |
| **(iv)** | Asiatic Lion is an example of invasive species in India. | **False** | The Asiatic Lion (*Panthera leo persica*) is an **endemic and endangered species** native to India (found only in Gir Forest, Gujarat). It is **not an invasive species**. |
| **(v)** | The National Green Tribunal (NGT) was set up under the NGT Act in the year 2010. | **True** | The National Green Tribunal Act was passed in **2010**, and the NGT was formally established on **October 18, 2010**. |

**You Asked**

(b) Fill in the blanks : (5×1=5)

(i) Ozone is found in the ________ layer of atmosphere

(ii) There are currently ________ biodiversity hotspots across the world.

(iii) Photochemical Smog formation occurs mostly during ________ season.

(iv) An international agreement signed in 1987, for phasing out ozone depleting substances is ________ ________.

(v) ________ is an example of invasive species in India.

Here are the answers to the fill-in-the-blanks:

| | Answer |
|---|---|
| **(i)** | Ozone is found in the **stratosphere** layer of atmosphere. |
| **(ii)** | There are currently **36** biodiversity hotspots across the world. |
| **(iii)** | Photochemical Smog formation occurs mostly during **summer** season. |
| **(iv)** | An international agreement signed in 1987, for phasing out ozone depleting substances is **Montreal Protocol**. |
| **(v)** | **Lantana camara** (or *Parthenium hysterophorus* / Water hyacinth / *Prosopis juliflora*) is an example of invasive species in India. |

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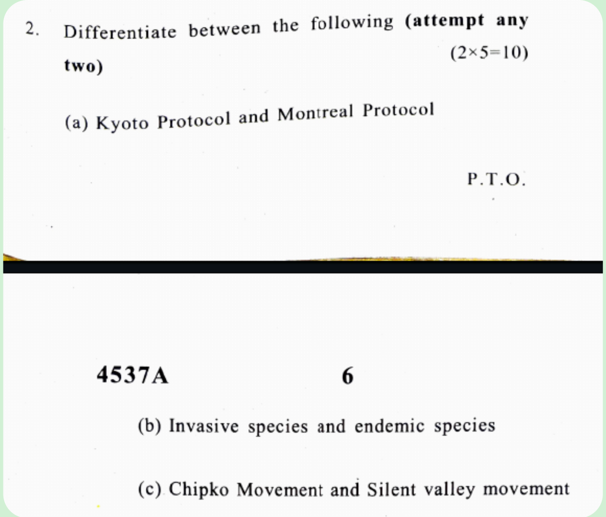
## Brief Explanations:

- **Stratosphere** — The ozone layer is located approximately 10–50 km above Earth's surface in the stratosphere, where UV radiation converts oxygen ($O_2$) into ozone ($O_3$).
- **36 biodiversity hotspots** — Conservation International recognizes 36 hotspots globally; India contains parts of two: the **Himalayas** and the **Western Ghats-Sri Lanka** hotspot.
- **Summer season** — Photochemical smog forms when sunlight (UV radiation) reacts with nitrogen oxides ($NO_x$) and volatile organic compounds (VOCs) in the atmosphere. It is most prevalent during **summer** when sunlight is intense and temperatures are high.
- **Montreal Protocol** — Signed on September 16, 1987, it is the most successful environmental treaty, phasing out CFCs, halons, and other ozone-depleting substances (ODS). September 16 is celebrated as the **International Day for the Preservation of the Ozone Layer**.
- **Invasive species in India** — **Lantana camara** is one of the most problematic invasive plants, spreading rapidly across Indian forests and displacing native vegetation. Other examples include **Parthenium** (Congress grass), **Water hyacinth** (clogs water bodies), and **Prosopis juliflora** (Vilayati babul).

You Asked


2. Differentiate between the following (attempt any two) (2×5=10)

(a) Kyoto Protocol and Montreal Protocol

P.T.O.

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(b) Invasive species and endemic species

(c) Chipko Movement and Silent valley movement

Kimi

Here are the differentiations for all three pairs — you may choose any two:

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### (a) Kyoto Protocol vs Montreal Protocol

| Basis | Kyoto Protocol | Montreal Protocol |
|---|---|---|
| **Year Adopted** | 1997 (entered into force in 2005). | 1987 (entered into force in 1989). |
| **Target Problem** | **Global warming and climate change** caused by greenhouse gas (GHG) emissions. | **Ozone layer depletion** caused by ozone-depleting substances (ODS). |
| **Target Gases** | **Greenhouse gases**: $CO_2$, $CH_4$, $N_2O$, HFCs, PFCs, $SF_6$. | **Ozone-depleting substances**: CFCs, halons, carbon tetrachloride, methyl chloroform, HCFCs. |
| **Binding Commitments** | Legally binding emission reduction targets **only for developed (Annex I) countries**. | Binding phase-out schedules for **all signatory countries**, with differentiated timelines for developed and developing nations. |
| **Mechanisms** | **Clean Development Mechanism (CDM)**, Joint Implementation (JI), Emissions Trading. | No market mechanisms; focused on technology transfer and financial assistance for phase-out. |
| **Success** | Mixed success; USA never ratified; Canada withdrew. Superseded by Paris Agreement (2015). | Most successful environmental treaty; ozone layer recovering, projected to return to 1980 levels by mid-century. |
| **India's Role** | India ratified but had no binding targets as a non-Annex I country. | India ratified and successfully phased out CFCs and halons ahead of schedule. |

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### (b) Invasive Species vs Endemic Species

| Basis | Invasive Species | Endemic Species |
|---|---|---|
| **Definition** | A non-native species introduced to an area that s preads rapidly, causing ecological or economic ha rm. | A species that is **native to and restricted to** a spec ific geographic area and found nowhere else on Ea rth. |
| **Origin** | **Introduced** from outside the native range — accid entally or deliberately. | **Native** to the region; evolved there over long period s. |
| **Impact on Ecosystem** | Often **harmful** — outcompetes native species, dis rupts food webs, alters habitats, causes extinctio ns. | **Integral** to the ecosystem — part of natural biodive rsity, often with co-evolved relationships. |
| **Population Trend** | Typically shows **explosive growth** due to lack of natural predators/competitors. | Usually **stable** but vulnerable to habitat loss and cli mate change. |
| **Conservati on Status** | Generally targeted for **eradication or control**. | Generally targeted for **protection and conservation**. |
| **Examples in India** | **Lantana camara, Parthenium hysterophorus, Water hyacinth, Prosopis juliflora, African catfish.** | **Lion-tailed Macaque, Nilgiri Tahr, Purple Frog, Asiatic Lion, Great Indian Bustard.** |
| **Key Concern** | Biodiversity loss, economic damage, human healt h risks. | Habitat destruction, small population size, high ext inction risk. |

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### (c) Chipko Movement vs Silent Valley Movement

| Basis | Chipko Movement | Silent Valley Movement |
|---|---|---|
| **Locatio n** | **Uttarakhand** (then Uttar Pradesh), Himalayan region. | **Kerala**, Western Ghats. |
| **Period** | **1973–1980s** (peak in 1970s). | **1970s–1983** (active resistance from late 197 0s). |
| **Main Issue** | **Commercial felling of trees** by contractors; threat to local forest-based livelihoods. | Proposed **Silent Valley Hydroelectric Project** threatening pristine tropical rainforest. |
| **Key Leader s** | **Sunderlal Bahuguna, Chandi Prasad Bhatt, Gaura Devi.** | **Sugathakumari, K. M. Chinnappa, environmental groups like KSSP (Kerala Sasthra Sahithya Parishad).** |
| **Method of Protest** | **Tree-hugging (chipko)** — villagers physically embraced tre es to prevent felling. | Scientific advocacy, public awareness campa igns, petitions, and political lobbying. |
| **Primar y Focus** | **Community rights over forests** and sustainable use of for est resources. | **Ecological/biodiversity conservation** — prote ction of endemic species like the Lion-tailed Macaque. |
| **Outco me** | Led to a **15-year ban on green felling** in Himalayan forests above 1,000 meters (1980). Influenced India's social fores try and joint forest management policies. | Project was **abandoned in 1983**; Silent Valley was declared a **National Park in 1984** and late r a UNESCO World Heritage Site. |

| Basis | Chipko Movement | Silent Valley Movement |
|---|---|---|
| **Legacy** | Symbol of **community-based environmentalism** and non-violent resistance globally. | Pioneered **science-based environmental activism** in India; demonstrated the power of informed public campaigns. |

You Asked

3. Write short notes on the following (**attempt any two**) (2×5=10)

(a) Environmental legislation in India

(b) Resettlement and rehabilitation of people affected by any developmental project

(c) Clean energy alternatives to combat global warming

Kimi

Here are short notes on all three topics — you may choose any two:

---

## (a) Environmental Legislation in India

India has developed a comprehensive framework of environmental laws to protect natural resources, control pollution, and ensure sustainable development. Key legislations include:

### 1. The Wildlife (Protection) Act, 1972

- India's umbrella law for wildlife conservation.
- Established protected areas: **National Parks, Wildlife Sanctuaries, and Tiger Reserves**.
- Schedules species into protection categories; prohibits hunting of endangered species.
- Led to the creation of **Project Tiger (1973)** and **Project Elephant**.

### 2. The Water (Prevention and Control of Pollution) Act, 1974

- First major environmental legislation in India.
- Established **Central and State Pollution Control Boards (CPCB/SPCBs)**.
- Regulates discharge of pollutants into water bodies.

### 3. The Air (Prevention and Control of Pollution) Act, 1981

- Empowers CPCB/SPCBs to monitor and control air pollution.

- Sets emission standards for industries and vehicles.

### 4. The Environment (Protection) Act, 1986

- Enacted after the **Bhopal Gas Tragedy (1984)**.
- Umbrella legislation empowering the Central Government to take measures for environmental protection.
- Parent Act for numerous rules: **Hazardous Waste, Biomedical Waste, Plastic Waste, E-Waste, and CRZ Notifications**.

### 5. The National Green Tribunal Act, 2010

- Established the **National Green Tribunal (NGT)** as a quasi-judicial body.
- Adjudicates environmental disputes and enforces environmental laws.

### 6. The Forest (Conservation) Act, 1980

- Restricts diversion of forest land for non-forest purposes.
- Requires Central Government approval for any de-reservation of forests.

### 7. The Biological Diversity Act, 2002

- Implements India's obligations under the **Convention on Biological Diversity (CBD)**.
- Established the **National Biodiversity Authority (NBA)** to regulate access to biological resources.

### 8. The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA)

- Recognizes **individual and community forest rights** of tribal and forest-dwelling communities.
- Grants **gram sabha** authority over forest management and displacement decisions.

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## (b) Resettlement and Rehabilitation of People Affected by Developmental Projects

Developmental projects (dams, mines, highways, industries) often displace local communities, necessitating comprehensive resettlement and rehabilitation (R&R) frameworks.

### Key Challenges

- **Inadequate compensation** — Below-market rates; landless laborers often excluded.
- **Loss of livelihoods** — Farmers lose land; wage laborers lose local employment.
- **Social disintegration** — Breakdown of community bonds, kinship networks, and cultural identity.
- **Poor resettlement sites** — Lack of water, sanitation, schools, and healthcare.
- **Delayed implementation** — Bureaucratic hurdles and weak enforcement.

### Legal and Policy Frameworks

| Framework | Key Provisions |
|---|---|
| **RFCTLARR Act, 2013** | Replaced the 1894 Act; mandates **Social Impact Assessment (SIA)**, **informed consent**, compensation up to **4x market value** in rural areas, and R&R entitlements for all affected families. |
| **National Rehabilitation and Resettlement Policy, 2007** | First comprehensive national framework; emphasized minimum displacement and community participation. |

| Framework | Key Provisions |
|---|---|
| **FRA, 2006** | Requires **gram sabha consent** for displacement of tribal communities; recognizes forest rights. |
| **PESA, 1996** | Grants autonomy to gram sabhas in Schedule V areas to approve/reject land acquisition. |

### Best Practices and Recommendations

- **Participatory planning** involving affected communities from the outset.
- **Livelihood restoration** through skill training, employment guarantees, and alternative income sources.
- **Culturally appropriate resettlement** respecting tribal customs and sacred sites.
- **Long-term monitoring** of resettled communities' well-being.
- **Gender-sensitive approaches** ensuring women are included in compensation and decision-making.

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## (c) Clean Energy Alternatives to Combat Global Warming

Global warming, driven by fossil fuel combustion, requires a transition to **clean energy alternatives** that produce minimal or zero greenhouse gas emissions.

### 1. Solar Energy

- **Photovoltaic (PV) cells** convert sunlight directly into electricity.
- **Solar thermal** systems use sunlight to generate heat for power generation.
- India has the **National Solar Mission** (JNNSM) with a target of **100 GW solar capacity**.
- **Advantages:** Abundant, renewable, declining costs, minimal water use.
- **Challenges:** Intermittency, land requirements, storage needs.

### 2. Wind Energy

- Wind turbines convert kinetic energy of wind into electricity.
- India is the **4th largest wind power producer** globally.
- **Advantages:** Mature technology, cost-competitive, no fuel costs.
- **Challenges:** Site-specific, intermittency, impact on birds and bats.

### 3. Hydropower

- Uses flowing water to generate electricity; includes **run-of-river** and **pumped storage** systems.
- India has significant untapped potential in the **Himalayas and North-East**.
- **Advantages:** Reliable baseload power, long lifespan.
- **Challenges:** Environmental and social impacts of large dams, sedimentation.

### 4. Biomass and Bioenergy

- Uses organic matter (agricultural waste, forest residues, biogas) for energy.
- **Biogas plants** convert animal dung and organic waste into methane for cooking and electricity.
- **Advantages:** Utilizes waste, rural employment, carbon-neutral if sustainably managed.
- **Challenges:** Land-use competition with food crops, air quality concerns.

### 5. Geothermal Energy

- Harnesses heat from Earth's interior for electricity and direct heating.
- Limited potential in India but viable in **Ladakh and Himalayan regions**.
- **Advantages:** Baseload capacity, small land footprint.
- **Challenges:** High upfront costs, location-specific.

### 6. Nuclear Energy

- Uses nuclear fission to generate heat and electricity.
- India operates **22 nuclear reactors** with plans for expansion.
- **Advantages:** Low carbon emissions, high energy density, reliable baseload.
- **Challenges:** Radioactive waste management, safety concerns, high capital costs.

### 7. Green Hydrogen

- Hydrogen produced using renewable energy through electrolysis of water.
- Can replace fossil fuels in **industry, transport, and power**.
- India launched the **National Hydrogen Mission** to become a global hub.

### Complementary Strategies

- **Energy efficiency** — Reducing demand through better buildings, appliances, and industrial processes.
- **Grid modernization** — Smart grids, energy storage (batteries, pumped hydro), and demand response.
- **Electrification of transport** — Electric vehicles powered by renewable grids.

You Asked

4. (a) Give a brief account of Project Tiger.

(b) Discuss the importance of the Tiger as a flagship species for wildlife conservation in India.

(5+5)

Kimi

Here are comprehensive answers to both parts:

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## 4. (a) Brief Account of Project Tiger

**Project Tiger** is India's flagship species conservation programme, launched on **April 1, 1972**, in response to the alarming decline in tiger populations due to poaching, habitat loss, and human-wildlife conflict. It was initiated at **Jim Corbett National Park** in Uttarakhand.

### Objectives

- Ensure a viable population of Bengal tigers in their natural habitats.
- Preserve areas of biological importance as a natural heritage for future generations.
- Maintain tiger populations for ecological, scientific, economic, aesthetic, and cultural values.

### Key Features

| Feature | Description |
|---|---|
| **Tiger Reserves** | Designated protected areas with a **core/critical tiger habitat** (strictly protected) and a **buffer zone** (limited human activity). India now has **54 tiger reserves** covering approximately 78,000 km². |
| **NTCA** | The **National Tiger Conservation Authority** was constituted in **2006** under the Wildlife (Protection) Act amendment to provide statutory backing and oversee Project Tiger. |
| **Monitoring** | Uses **M-STrIPES** (Monitoring System for Tigers — Intensive Protection and Ecological Status), a camera trap-based system for population estimation. |
| **All-India Tiger Estimation** | Conducted every **4 years**; the 2022 census recorded **3,682 tigers** — the highest globally. |
| **Funding** | Centrally Sponsored Scheme with funding from the Ministry of Environment, Forest and Climate Change (MoEFCC). |

### Successes

- India is home to **over 75% of the world's wild tigers**.
- Tiger numbers have risen from **1,411 (2006)** to **3,682 (2022)**.
- Project Tiger is considered one of the **most successful species conservation programmes** globally.
- It has indirectly protected vast forest ecosystems and countless other species.

### Challenges

- Poaching and illegal wildlife trade.
- Human-wildlife conflict in buffer zones.
- Habitat fragmentation and linear infrastructure (roads, railways).
- Prey depletion in some reserves.

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## 4. (b) Importance of the Tiger as a Flagship Species for Wildlife Conservation in India

A **flagship species** is a charismatic, high-profile species chosen to represent an environmental cause and mobilize public support and funding for conservation. The **Royal Bengal Tiger** serves as India's most powerful flagship species.

### 1. Ecological Significance

- The tiger is an **apex predator** at the top of the food chain.
- Its presence indicates a **healthy, functioning ecosystem** with sufficient prey base and intact habitat.
- Protecting tigers automatically protects entire ecosystems — a phenomenon known as the **"tiger landscape" approach**.

### 2. Umbrella Species Effect

- Tigers require **large home ranges** (up to 100 km² per individual).
- Protecting tiger habitat under Project Tiger conserves vast tracts of forest that also shelter **elephants, leopards, deer, wild boar, birds, reptiles, and innumerable plant species**.
- Example: **Corbett Tiger Reserve** protects over 500 species of plants and 600 species of animals.

### 3. Economic and Tourism Value

- Tiger reserves are major **eco-tourism destinations**, generating revenue for local communities and states.
- **Ranthambore, Bandhavgarh, and Kanha** attract millions of tourists annually, creating employment and incentivizing conservation.

### 4. Cultural and National Identity

- The tiger is India's **national animal** and a symbol of strength and grace.
- It features prominently in Indian mythology, folklore, and art.
- This cultural connection makes the tiger a powerful **symbol for conservation awareness**.

### 5. Mobilization of Resources and Political Will

- The tiger's charisma helps **attract international funding** (e.g., Global Tiger Forum, WWF, NTCA partnerships).
- It galvanizes **political will** for forest protection and anti-poaching measures.
- India's hosting of the **Global Tiger Summit** and commitment to doubling tiger numbers (Tx2 goal) demonstrate this.

### 6. Catalyst for Policy and Legal Frameworks

- Project Tiger led to the creation of the **NTCA** and strengthened the **Wildlife (Protection) Act**.
- It drives **anti-poaching technology** (camera traps, drones, patrolling) and **corridor protection** policies.

### 7. International Prestige

- India's success in tiger conservation enhances its **global environmental leadership**.
- It demonstrates India's commitment to **SDG 15 (Life on Land)** and the **Convention on Biological Diversity (CBD)**.

### Conclusion

The tiger is not merely a species to be saved — it is a **conservation catalyst**. As a flagship species, it channels public emotion, scientific effort, policy attention, and financial resources toward protecting India's natural heritage. Saving the tiger means saving India's forests, rivers, wildlife, and ultimately, the ecological security of the nation.

You Asked

5. Environmental movements in India have played a significant role in Biodiversity conservation and environmental protection. Justify this statement with appropriate examples. (10)

Here is a comprehensive answer to the question on **Environmental Movements in India**:

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## Environmental Movements in India: Significant Role in Biodiversity Conservation and Environmental Protection

### Introduction

India's environmental movements represent some of the most powerful examples of grassroots activism in the world. Emerging from the intersection of ecological concerns, social justice, and indigenous rights, these movements have fundamentally shaped India's conservation policies, legal frameworks, and public consciousness. They demonstrate that environmental protection in India is not merely a top-down government exercise but a deeply democratic, community-driven process.

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### 1. Chipko Movement (1973–1980s) — Uttarakhand

**Background:** Commercial logging by contractors threatened the livelihoods of Himalayan villagers dependent on forests for fuel, fodder, and minor forest products.

**Action:** Led by **Sunderlal Bahuguna, Chandi Prasad Bhatt, and Gaura Devi**, villagers embraced trees (*chipko* means "to hug") to prevent felling.

**Impact on Conservation:**

- Led to a **15-year ban on green felling** above 1,000 meters in the Himalayas (1980).
- Influenced India's **Social Forestry** and **Joint Forest Management (JFM)** policies.
- Established the principle of **community rights over forests**.
- Became a global symbol of non-violent environmental resistance.

**Biodiversity Protection:** Saved thousands of hectares of Himalayan forests that harbor endemic species like the **Himalayan Monal, Snow Leopard, and diverse oak-rhododendron ecosystems**.

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### 2. Silent Valley Movement (1970s–1983) — Kerala

**Background:** The proposed **Silent Valley Hydroelectric Project** threatened to submerge 8,000 hectares of pristine tropical rainforest in the Western Ghats.

**Action:** Led by **Sugathakumari, K. M. Chinnappa, and KSSP (Kerala Sasthra Sahithya Parishad)**, the movement used scientific data, public awareness campaigns, and political lobbying.

**Impact on Conservation:**

- The project was **abandoned in 1983**.
- **Silent Valley was declared a National Park in 1984** and later became part of the **UNESCO Nilgiri Biosphere Reserve** and a **World Heritage Site**.
- Pioneered **science-based environmental activism** in India.

**Biodiversity Protection:** Protected one of the last undisturbed tropical rainforests, home to the **Lion-tailed Macaque, Nilgiri Langur, Malabar Giant Squirrel, and over 1,000 plant species**.

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### 3. Narmada Bachao Andolan (NBA) (1985–present) — Gujarat, Madhya Pradesh, Maharashtra

**Background:** The **Sardar Sarovar Dam** and other dams on the Narmada River threatened to displace over **200,000 people** and submerge vast tracts of forest and agricultural land.

**Action:** Led by **Medha Patkar, Baba Amte, and Arundhati Roy**, the movement used non-violent resistance, hunger strikes, and international advocacy.

**Impact on Conservation:**

- Forced the **World Bank to withdraw funding** in 1993 — a landmark victory.
- Raised global awareness about **development-induced displacement**.
- Influenced the **RFCTLARR Act, 2013** (Right to Fair Compensation and Transparency in Land Acquisition).
- Led to improved **environmental impact assessment** requirements for dams.

**Biodiversity Protection:** Protected riparian forests, aquatic ecosystems, and the Narmada's unique biodiversity, including the **Gharial, Indian Skimmer, and numerous fish species**.

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### 4. Appiko Movement (1983) — Karnataka

**Background:** Inspired by Chipko, this movement responded to commercial felling in the **Uttara Kannada district** of the Western Ghats.

**Action:** Led by **Pandurang Hegde**, protesters hugged trees, organized padayatras, and used folk performances.

**Impact on Conservation:**

- Halted commercial felling in several forest areas.
- Promoted **sustainable forest management** and community forestry.
- Strengthened the Western Ghats conservation discourse.

**Biodiversity Protection:** Protected Western Ghats forests rich in **endemic amphibians, reptiles, and plant species**.

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### 5. Bishnoi Movement (1730) — Rajasthan

**Background:** The earliest recorded environmental movement in India. **363 Bishnois** led by **Amrita Devi** sacrificed their lives to protect **Khejri trees** from being cut for the Maharaja of Jodhpur's palace.

**Impact on Conservation:**

- The Maharaja decreed that no green tree would be cut and no animal hunted in Bishnoi areas.
- Established the **Bishnoi community** as lifelong environmental stewards.
- Influenced modern conservation ethics and the **Wildlife (Protection) Act, 1972**.

**Biodiversity Protection:** The Bishnois continue to protect the **Blackbuck, Chinkara, Khejri tree, and numerous desert species** in Rajasthan.

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### 6. Save the Western Ghats Movement (1980s)

**Background:** The Western Ghats — a global biodiversity hotspot — faced threats from mining, dams, and deforestation.

**Action:** A coalition of scientists, activists, and local communities campaigned for protection.

**Impact on Conservation:**

- Led to the **Gadgil Committee (2011)** and **Kasturirangan Committee (2013)** reports on Western Ghats ecology.
- Influenced the designation of **Ecologically Sensitive Areas (ESAs)** in the Western Ghats.

**Biodiversity Protection:** The Western Ghats harbor **over 5,000 endemic plant species, 139 mammal species, 508 bird species, and 179 amphibian species** — many found nowhere else on Earth.

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### 7. Tehri Dam Movement (1970s–2000s) — Uttarakhand

**Background:** The **Tehri Dam** on the Bhagirathi River threatened to submerge the old town of Tehri and surrounding forests.

**Action:** Led by **Sunderlal Bahuguna**, the movement highlighted seismic risks and displacement.

**Impact on Conservation:**

- Delayed the project for decades and forced improved **environmental and seismic assessments**.
- Raised awareness about the **Himalayan ecosystem's fragility**.

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## Broader Impact of Environmental Movements

| Domain | Impact |
|---|---|
| **Policy** | Influenced the Wildlife (Protection) Act, FRA 2006, RFCTLARR Act 2013, NGT Act 2010. |
| **Legal** | Established the right to a clean environment as part of the **Right to Life (Article 21)**. |
| **Institutional** | Led to the creation of the **MoEFCC, NTCA, NGT, and State Biodiversity Boards**. |
| **Public Awareness** | Created a culture of environmental consciousness and citizen activism. |
| **International** | Positioned India as a leader in community-based conservation and sustainable development. |

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## Conclusion

Environmental movements in India have been instrumental in **shifting the narrative from development-at-all-costs to sustainable, inclusive development**. They have saved forests, rivers, and species; empowered marginalized communities; and created robust legal and institutional frameworks for conservation. From the Bishnoi martyrs of 1730 to the Narmada activists of today, these movements prove that **environmental protection is inseparable from social justice** — and that India's biodiversity heritage owes much to the courage and conviction of its environmental activists.